

جامعة الخرطوم



كلية الهندسة – قسم الهندسة الكهربائية والإلكترونية

امتحانات الملاحق والبدائل الصف الثالث لنيل درجة بكالوريوس الشرف في الهندسة الكهربائية والإلكترونية

الفصل الدراسي الثاني للعام الدراسي 2023 - 2024

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عدد الصفحات 3

التاريخ: ديسمبر 2025

**Question 1:** briefly answer the following questions.

- 1.1 What do we mean by saying that Java is architecture-neutral language?
- 1.2 How did the World Wide Web and the internet participate in the programming revolution?
- 1.3 What is the main difference between OOP and POP?
- 1.4 What are the components of the JRE?
- 1.5 What is the difference between inheritance and polymorphism?
- 1.6 What do we mean by encapsulation?
- 1.7 There are two ways to achieve abstraction. Mention them.
- 1.8 A thread could be terminated due to three reasons. Mention them.
- 1.9 What are the three default streams in Java?
- 1.10 On which API Java swing was built on top of?

**Question 2:** state if the sentence is true or false and correct the false statements.

**2.1** The private modifier is used when we want the code to be accessible only within the same declared class.

**2.2** Run time polymorphism is called static polymorphism.

**2.3** The Java byte code is a .class file that is executed by the Java run-time system.

**2.4** The concept of multiple inheritance is widely used in Java.

**2.5** An interface can inherit as many as other interfaces.

**2.6** Errors could arise due to a user non valid input.

**2.7** There is one JDBC driver for all database sources.

**Question 3:** fill in the blanks.

**3.1** \_\_\_\_\_ is a Java API used to connect to and execute queries against databases, allowing Java applications to interact with structured data.

**3.2** A key advantage of Java threads is that they are \_\_\_\_\_, meaning they share the same memory space and can easily communicate.

**3.3** . In Java, a class can implement multiple \_\_\_\_\_

**3.4** a \_\_\_\_\_ is a component which can contain other elements like buttons.

**3.5** The model- view- \_\_\_\_\_ architecture is used in Swing.

**3.6** Examples of procedural oriented programming languages are \_\_\_\_\_ programming language.

**3.7** In Java polymorphism has two types: Compile-time Polymorphism and \_\_\_\_\_ Polymorphism.

**Question 4:** For each block code, complete the output and provide explanation for your answer:

#### 4.1

```
public class Exams {  
    public static void main(String[] args) {  
  
        String[] seats = {"M", "C", "G", "P", "D"};  
        System.out.println("It's time to take your 3th years exams. Please, take your seats.\n");  
        for (int i = 0; i < seats.length; i++) {  
            System.out.println(CODE);  
        }  
    }  
}
```

**Output:**

It's time to take your 3th years exams. Please, take your seats.

M, you will take seat 0  
C, you will take seat 1  
G, you will take seat 2  
P, you will take seat 3  
D, you will take seat 4

#### 4.2

```
public class Agree {  
    public static void main(String[] args) {  
        int totalVotes = 760;  
        System.out.println("Statement: If Java was easy, they would call it Python!");  
        CODE  
    }  
    public static void calculateVotes(double numOfVotes) {  
        // 15% of votes do agree with the statement.  
  
        double portion = numOfVotes*0.15;  
  
        System.out.println("The number of people who agree with this statement is " + (int) portion);  
    }  
}
```

**Output:**

Statement: If Java was easy, they would call it Python!  
The number of people who agree with this statement is 114